TRUTHLENS AI

# Frontend Design & Architecture Guide

A source-grounded handoff for independent UI/UX, visual design, accessibility, and frontend architecture review.

Prepared from the repository implementation on 27 October 2026

## PRODUCT

An Indian financial claim and forwarded-message analyzer. The interface separates evidence-based factual status from heuristic scam-risk assessment, and exposes the sources and pipeline details behind each result.

## REVIEW TARGET

Use this brief with Claude, Gemini, or a human design reviewer. Recommendations should respect uncertainty, evidence provenance, and user safety while making the experience clearer and more polished.

# How to use this handoff

This document describes the frontend as implemented, not a proposed redesign. Technical details and token values come from the checked-in source files.

Share the PDF with a reviewer and ask them to assess the experience against the product goals and constraints below. Ask for concrete, prioritized UI changes with examples, not an abstract redesign. The final section contains a ready-to-paste review prompt.

| Review section | What it covers |
|---|---|
| 01 - Product and user journey | Entry screen, analysis state, results view, trust and safety goals. |
| 02 - Frontend stack | Runtime, framework, build chain, dependencies, API boundary, project structure. |
| 03 - Screens and components | Component responsibilities and page composition. |
| 04 - Visual system | Exact CSS tokens, typography, surface treatment, status colors, iconography. |
| 05 - Layout and interaction | Responsive layout, tabs, filters, loading, validation, keyboard behavior. |
| 06 - Motion and accessibility | Actual animation and transitions, semantics, keyboard and contrast review. |
| 07 - Data model and trust boundaries | What the UI receives, what each result means, and what it must not imply. |
| 08 - Opportunity list | Specific design-review prompts and suggested priority order. |

## Executive snapshot

The current frontend is a dark, card-based React application with a dedicated claim-entry screen and a separate results screen. Results are organized into Overview, Evidence, and Diagnostics tabs. The interface has strong auditability and careful uncertainty language, but its density and repeated technical terminology can make the main decision harder to scan. Reviewers should improve hierarchy and comprehension without merging factual truth with scam likelihood.

**Core product distinction: factual verdict answers whether available evidence supports or contradicts a claim; scam risk answers whether behavioral warning patterns appear in the submitted message. A high scam-risk score is not proof that the factual claim is false, and a low score is not proof that a message is safe.**

# 01. Product and user journey

The main journey has a focused input state, an in-progress state, and an evidence-oriented result state.

| State | Current UI behavior | Design goal |
|---|---|---|
| Entry | Brand header and backend status; short intro; analyzer card; four sample scenarios; textarea; character count; Analyze Claim button; capability cards. | Make the next action obvious and set expectations about evidence and safety. |
| Loading | Brain icon/pulse, three pipeline labels: Extract claims, Search evidence, Assess risk. Copy says it can take a few seconds. | Show progress without suggesting a step has finished when exact progress is unavailable. |
| Results | New analysis action, request ID, evidence coverage status, submitted message, three metrics, then Overview / Evidence / Diagnostics tabs. | Put the takeaway first; make evidence inspectable; keep implementation diagnostics available without dominating. |
| Error / offline | Backend health banner with retry; analysis failure banner preserves submitted text and offers retry. | Explain recovery clearly and preserve user work. |

The entry screen is not a generic fact-checker: its copy explicitly targets financial claims, WhatsApp forwards, KYC alerts, and investment offers in India. Four demo inputs represent an account-block fee threat, DICGC deposit insurance, guaranteed-return scheme, and KYC phishing link.

## Results overview

- Verdict panel: TRUE, FALSE, PARTIALLY_TRUE, UNVERIFIED, or ERROR, with an evidence-grounded explanation.
- Independent scam-risk panel: LOW, MEDIUM, or HIGH category; 0-100 heuristic score; thresholds returned by backend; matched indicators and explanation.
- Extracted assertions: decomposed statements, assertion type, and entities.

## Evidence tab

- Relevant, Context, and All results filters, with counts; source quality, provider, page-versus-snippet provenance, date, assertion relationship, excerpt, source URL, and retrieval note.

## Diagnostics tab

- Expandable pipeline audit showing request/status, evidence retrieval outcome, providers attempted, source/page/snippet counts, assertion coverage, targeted queries, limitations, and processing notes.

# 02. Frontend stack and architecture

Version strings below are the declared package ranges in frontend/package.json, not a lockfile export.

| Layer | Implementation | Notes |
|---|---|---|
| UI runtime | React 19.3.1, React DOM 19.3.1 | Functional components and React hooks; StrictMode entry. |
| Language | TypeScript ~6.0.3 | Strict app compiler settings include noUnusedLocals and noUnusedParameters. |
| Dev server/build | Vite 8.3.4, @vitejs/plugin-react 6.1.2 | Vite serves the app; build script runs tsc -b then vite build. |
| Icons | lucide-react ^1.61.0 | Line icons used for brand, status, navigation, evidence, verdict, and risk indicators. |
| Lint | oxlint ^1.88.0 | npm run lint runs oxlint. |
| Styling | Plain CSS | Global tokens in src/index.css; app/components and page styles in src/App.css. No component library or CSS framework declared. |
| Fonts | Inter + JetBrains Mono via Google Fonts | Inter for UI; JetBrains Mono for code, IDs, and technical labels, with system fallbacks. |

## Frontend source map

```
frontend/
  index.html                document metadata + font links
  src/main.tsx              React StrictMode entry
  src/App.tsx               application state, page flow, result tabs
  src/App.css               component and page styling
  src/index.css             global reset, design tokens, base elements
  src/types/index.ts        API-aligned frontend data contracts
  src/services/api.ts       health/analyze HTTP client
  src/components/           Header, analyzer, result panels, footer
  package.json              scripts and dependencies
```

## API boundary

The frontend reads VITE_API_BASE_URL (default http://127.0.0.1:8000), calls GET /api/health and POST /api/analyze, validates basic response fields, and converts network, HTTP, timeout, malformed JSON, and incomplete-schema cases into user-facing errors. The analysis client sets a 25-second timeout; health uses 5 seconds. The backend endpoint is the source of analysis content; the frontend presents rather than independently verifies it.

## Build commands

```
cd frontend
npm run dev      # local Vite development server
npm run build    # TypeScript project build + Vite production bundle
npm run lint     # Oxlint
```

# 03. Screen and component inventory

Responsibilities and high-value review questions for each visible area.

| Component / file | What it renders | Review focus |
|---|---|---|
| App - src/App.tsx | Health polling every 30 seconds; analysis and retry state; entry/loading/results composition; result tabs; evidence metrics. | State transitions, page hierarchy, stale result behavior, tab semantics, and how key conclusions are surfaced. |
| Header - components/Header.tsx | ShieldCheck brand, product subtitle, backend connection pill, manual refresh. | Brand scale, status wording, and whether backend availability is distinct from verification quality. |
| AnalysisWorkspace.tsx | Demo chips; textarea; clear button; validation; 5,000-character limit; Ctrl/Cmd+Enter submit; loading button. | Demo discoverability, input affordance, scenario labels, and mobile wrapping. |
| VerdictPanel.tsx | Factual outcome, icon/status styling, explanation, extracted-assertion count, independent high/medium/low scam safety callout for UNVERIFIED. | Make epistemic status and safety action legible without implying scam risk proves falsity. |
| ScamRiskPanel.tsx | Heuristic score, category meter, threshold labels, summary, indicators/weights, disclaimer. | Avoid probability-like interpretation of a heuristic; show why a score was assigned. |
| AssertionsPanel.tsx | Assertion cards with number, text, type, entities. | Help a reader understand which statement is actually being assessed. |
| EvidencePanel.tsx | Relevant/context/all source views, quality/provenance badges, assessments, excerpt, URL, errors. | Readable source cards, freshness, evidence relevance, and distinction between snippet and fetched page. |
| DiagnosticsPanel.tsx | Expandable operational details and coverage/query audit. | Keyboard interaction, progressive disclosure, accessible disclosure semantics. |
| Footer.tsx | Responsible-use / regulatory disclaimer and links to RBI, SEBI, cybercrime.gov.in, 1930. | Keep legally important guidance readable without turning the footer into a wall of text. |

## Icon system

Lucide React outline icons are used consistently for shield/identity, status, search, scales, help, alerts, book/evidence, terminal/diagnostics, and external links. Meaning is generally accompanied by text labels, which is preferable to color or icon-only communication.

Current text/data styling uses monospaced typography for request IDs, evidence snippets in some contexts, matched phrases, search queries, and technical statuses. Reviewers should make sure monospace is reserved for genuinely technical or literal text so it does not make large user-submitted messages feel harder to read.

# 04. Visual system and color grading

Exact values come from frontend/src/index.css. The report shows opaque visual approximations for alpha colors.

| Token | Value | Role |
|---|---|---|
| --bg-app | #080C14 | Application backdrop; near-black navy. |
| --bg-surface | #0F172A | Primary cards and surface. |
| --bg-surface-elevated | #1E293B | Raised cards, chips, and nested surfaces. |
| --bg-surface-hover | #273549 | Interactive hover surface. |
| --accent-cyan | #38BDF8 | Primary brand, links, focus border, action emphasis. |
| --accent-indigo | #6366F1 | Secondary brand accent. |
| --text-main / secondary / muted | #F8FAFC / #94A3B8 / #64748B | Primary reading, supporting text, low-emphasis labels. |
| --color-true | #10B981 | Positive/confirmed state. |
| --color-false | #EF4444 | False/debunked state and danger. |
| --color-partial | #F59E0B | Partial truth and caution. |
| --color-unverified | #818CF8 | Unknown / inconclusive state. |

## Companion token groups

Subtle and status backgrounds use RGBA variants: brand cyan and indigo at 12% opacity; true/false/partial/unverified fills at 10% and borders at 30%. Borders are white at 8% or 14%, with a cyan focus border. Shadow levels are dark black at 30%, 40%, and 50%. Radius tokens are 6px, 10px, 14px, and pill/full 9999px.

## Palette read

The visual tone is dark navy / midnight slate, with cyan as the primary action and link color, indigo for unverified/secondary emphasis, and semantic green/red/amber for outcome states. Cards use layered navy surfaces, subtle translucent borders, and restrained shadows. Some component styles also use direct hex values and gradients, so the CSS variables are not yet a complete single source of truth.

## Typography

| Use | Current family / behavior |
|---|---|
| UI and body | Inter, with -apple-system, BlinkMacSystemFont, Segoe UI, Roboto, Helvetica, Arial fallbacks. |
| Technical / literal | JetBrains Mono, with Fira Code, Consolas, Courier New fallbacks. |
| Scale | Component CSS uses rem values and responsive clamp() for the entry heading; global line-height is 1.5 with antialiased rendering. |

# 05. Layout, responsive behavior, and interactions

The current interface is responsive through CSS breakpoints; the app does not use a declared layout framework.

| Pattern | Current implementation |
|---|---|
| Main container | Maximum width 1180px, centered; page padding adjusted for small screens. |
| Entry heading | Centered up to 760px, responsive heading size with clamp; left-aligned on narrow screens. |
| Result overview | One column by default; verdict/risk grid becomes 1.15fr / 0.85fr at 900px. |
| Evidence cards | Evidence list becomes a two-column grid at 800px. |
| Capability cards | Two columns from 680px; mobile styles adjust header, content, and spacing at 640px. |
| Footer | Bottom-row content changes to horizontal from 640px. |

## Interaction inventory

- Sample chip fills the input; clear button empties it; textarea is capped at 5,000 characters and visually reports the count.
- Submit requires nonblank text; both Ctrl+Enter and Cmd+Enter submit from the textarea.
- Analysis disables inputs and changes the button to a running state; errors preserve input and provide retry.
- Header health check runs at startup, polls every 30 seconds, and can be manually refreshed.
- Result tabs are Overview, Evidence, Diagnostics; Evidence has Relevant, Context, All results filters.
- Diagnostics opens/closes by click, Enter, or Space; source links open a new tab with noopener/noreferrer.
- Status text and icons accompany color; evidence source quality and evidence origin are named with badges.

## Responsive review to request

Ask reviewers to check 320px-390px phone widths, 768px tablet widths, and 1280px+ desktop widths. In particular review long claim wrapping, filter and demo-chip overflow, two-column evidence readability, header status crowding, keyboard focus visibility, and whether tab content remains understandable when read linearly.

## Practical design check

**At any viewport, a first-time user should be able to answer within a few seconds: What is the factual verdict? Is there a separate scam warning? What evidence supports the assessment? What action should I take next? The safety actions should remain visible when factual verdict is UNVERIFIED.**

# 06. Motion, accessibility, and implementation details

Motion is intentionally concentrated in loading and small interaction feedback.

| Motion | Source behavior | Review note |
|---|---|---|
| Pulse ring | 2-second infinite cubic-bezier animation; grows and fades around the loader icon. | Avoid excessive attention capture; provide reduced-motion handling. |
| Spinner | 1-second linear infinite rotation; used during health checks and submit loading. | Should not be the only indication that work is happening. |
| Slow icon rotation | 8-second linear infinite rotation on the loading center icon. | Consider whether two simultaneous loader motions add value. |
| Risk meter | Width transition 0.6 seconds with an ease-out curve. | Respect reduced-motion preference and avoid implying precision beyond a heuristic score. |
| Controls / links | Mostly 0.15-0.2 second color, border, shadow, and transform transitions. | Keep focus feedback distinct from hover feedback. |

## Accessibility strengths visible in code

- Semantic header, main, section, article, footer, form, headings, and links are used.
- Form text area has an accessible label; validation and backend errors use role=alert; safety callout uses role=status.
- Result sections use tablist/tab/tabpanel roles and selected state; evidence filter buttons expose aria-pressed.
- Clear and refresh icon buttons have accessible labels; links opened in a new tab set rel=noopener noreferrer.
- Color-coded verdicts and risk signals also have textual labels.

## Accessibility items to review

- No prefers-reduced-motion rule was found in the inspected global/app CSS; add one for pulse/spin/meter transitions.
- Diagnostics uses a clickable div with role=button and manual keyboard support; a native button or details/summary disclosure would provide more reliable semantics.
- Tabs expose roles but should also support ArrowLeft/ArrowRight/Home/End and expected focus management if following the ARIA tabs pattern.
- Verify contrast for muted text (#64748B) against the dark surfaces and for all status/badge text at its actual size.
- Check focus-visible outlines, heading order, zoom/reflow at 200-400%, link target clarity, and screen-reader announcements when results arrive.
- Google Fonts are loaded remotely; consider self-hosting or defining a resilient offline fallback if reliable typography matters in a constrained demo environment.

# 07. Data contracts and trust boundaries

These boundaries are central to both the design and the product's credibility.

| Data object | UI-relevant fields | Display implication |
|---|---|---|
| AnalyzeResponse | request_id, status, original_text, claims, verdict, explanation, evidence, scam_assessment, limitations, processing_notes, evidence_diagnostics | One response contains independent factual, behavioral-risk, and audit information. |
| EvidenceItem | url, title, excerpt, published_date, source_quality, retrieval_success, evidence_origin, provider, assessments, fetch/retrieval errors | Label source quality and provenance. Search snippet is not equivalent to fetched page text. |
| EvidenceAssessment | assertion_text, relevance, relationship (supports/contradicts/conflicting/contextual/none), explanation | Relationship is per assertion/source; avoid implying that a source supports the entire message. |
| ScamAssessment | score, category, thresholds, indicators, explanation, disclaimer | Score is heuristic and behavioral; it is not a probability or factual verdict. |
| EvidenceDiagnostics | retrieval_outcome, providers, counts, assertion_coverage, targeted_queries | Communicate coverage and process failure separately from a claim being true/false. |

## Trust model for the interface

- Factual status requires evidence that directly supports or contradicts a decomposed assertion; when no qualifying evidence exists, UNVERIFIED is an honest, meaningful state.
- Scam heuristics can still warn about urgency, payment demand, transfer request, suspicious link, or guaranteed return when factual status is unverified.
- A zero relevant-source count, zero fetched pages, or snippet-only results are coverage facts and should be explicit; raw total search result count is not the same as useful evidence.
- Official regulatory links and contact instructions belong near consequential guidance, but should not be presented as an endorsement of this automated output.
- The result should distinguish 'search did not find evidence' from 'evidence shows the claim is false'.

## Known review opportunities (not claims of defects)

- The dark interface and many nested cards create a dense dashboard; test a more restrained hierarchy and reduce redundant labels.
- The verdict, safety warning, evidence quality, and next action compete for attention in the overview; prioritize the human decision path.
- The loading UI displays all pipeline labels as active at once; confirm if this accurately maps backend progress or should be framed as an illustrative sequence.
- A metric called relevant sources depends on assessment availability; clarify when counts are source-level versus assertion-level.
- Footer and feature-card copy may be long; assess progressive disclosure and concise wording.

# 08. Prioritized opportunities for an independent review

Ask reviewers to preserve the product's core logic while making the experience feel more polished, credible, and judge-ready.

| Priority | Area | Questions for the reviewer |
|---|---|---|
| P0 | Result hierarchy | Can the user immediately find factual verdict, separate scam risk, and one safe next action? Should the overview use a clearer 'What we found' summary before technical details? |
| P0 | Unverified states | How can the UI communicate 'unknown due to insufficient evidence' while still showing a strong independent scam warning when signals exist? What wording avoids conflating them? |
| P0 | Evidence trust | How should official/reputable/general sources, fetched-page text vs snippet, and supports/contradicts/context labels be shown so users do not over-trust weak evidence? |
| P1 | Density and hierarchy | Which cards, headings, metrics, or repeated labels can be removed or collapsed? What should be visible by default vs under Evidence/Diagnostics? |
| P1 | Visual refinement | Does the midnight navy/cyan/indigo palette feel polished and coherent? Improve spacing, type scale, surfaces, alignment, and accent use without losing status clarity. |
| P1 | Responsive and accessibility | Identify concrete mobile, keyboard, screen-reader, contrast, reduced-motion, focus, and 200% zoom improvements. |
| P2 | Judge presentation | What small interaction or visual detail would make the product memorable without adding decoration that weakens trust? |

## Copy-paste prompt for Claude or Gemini

**You are reviewing the attached implementation-grounded frontend brief for TruthLens AI, a React + TypeScript interface for Indian financial forwarded claims. Give a rigorous UI/UX and visual-design critique for a hackathon demo. Respect the distinction between the evidence-grounded factual verdict (TRUE/FALSE/PARTIALLY_TRUE/UNVERIFIED) and a separate heuristic scam-risk assessment. Do not recommend forcing a binary verdict when evidence is insufficient. Prioritize changes by P0/P1/P2, explain the user problem each solves, and propose concrete screen hierarchy, copy, layout, colors, typography, responsive behavior, accessibility and motion improvements. Focus on the current entry screen and Overview/Evidence/Diagnostics result tabs. Call out anything that may cause users to mistake a search snippet, heuristic score, or no-result state for proof. End with a concise redesign specification and a list of the 5 highest-impact changes. Avoid generic advice; tie every recommendation to facts in the brief.**

## Implementation files referenced

```
frontend/src/App.tsx
frontend/src/App.css
frontend/src/index.css
frontend/src/main.tsx
frontend/src/services/api.ts
frontend/src/types/index.ts
frontend/src/components/AnalysisWorkspace.tsx
frontend/src/components/Header.tsx
frontend/src/components/VerdictPanel.tsx
frontend/src/components/ScamRiskPanel.tsx
frontend/src/components/AssertionsPanel.tsx
frontend/src/components/EvidencePanel.tsx
frontend/src/components/DiagnosticsPanel.tsx
frontend/src/components/Footer.tsx
frontend/package.json
frontend/vite.config.ts
frontend/tsconfig.app.json
frontend/index.html
```

Scope note: This is a frontend design and architecture handoff, not a security audit, backend model audit, or guarantee of factual accuracy. Package versions are manifest declarations; browser appearance can vary with runtime font availability and viewport.